Conflicts surrounding the pharmacological management of acute agitation and violence: Balancing Medical Necessity and Human Rights in Psychiatric Care

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Abstract

Objective: This review examines the conflicts surrounding the pharmacological management of acute agitation and violence in psychiatric care, focusing on balancing medical necessity with respect for patients' human rights.

Methods: We conducted a comprehensive literature review of the subject. Recurring themes were identified from the literature and synthesized to ensure thorough coverage of related issues.

Results: Pharmacological interventions, referred to as chemical restraints, effectively mitigate immediate risks in this situation. However, their use raises significant ethical concerns, regarding patient autonomy and dignity. Human rights advocates strongly argue for a complete ban on chemical restraints, emphasizing their coercive nature. This stance has led to unwarranted criticism of medical staffs, potentially demoralizing those who aim to benefit patients. Nevertheless, chemical restraints remain a crucial component of patient management in cases of urgency. Concurrently, researchers are developing new pharmacological approaches to managing violence, which may offer potential solutions that address these ethical criticisms.

Conclusion: Balancing medical necessity with the protection of fundamental human rights is crucial in psychiatric ethics and clinical practice. This review examines the criticisms leveled at chemical restraints and their underlying rationale, exploring how to best integrate these concerns in the pursuit of alternatives. Further research is essential to develop strategies that optimize patient safety while enhancing autonomy and dignity.

Keywords

Acute agitation, Chemical restraint, Patient autonomy, Human rights, Coercive treatment

I. Introduction

Acute agitation and violence in psychiatric care present significant challenges for both patients and medical staffs caring for them. These incidents can rapidly escalate, leaving little time for management, potentially causing physical harm and disrupting the therapeutic environment. The management requires a delicate balance between ensuring immediate safety and respecting patients' rights, dignity and autonomy. This complexity sets the stage for the tumultuous debate surrounding the pharmacological management of acute agitation and violence in psychiatric care.

One of the effective measures to handle these incidents is often called "chemical restraint." It involves the use of medications, usually against the patient's will, to sedate them or limit their movement. This term often evokes highly negative connotations. Critics raise several concerns, including the potential for misuse or over-reliance on invasive measure when less restrictive options might suffice. Complications and other safety issues are also a concern. However, the most contentious aspect is the ethical questions surrounding patients' dignity and autonomy.

At the heart of this debate lies the question of how far one can infringe upon patients' human rights and autonomy for the sake of medical necessity. Admittedly, the currently available methods of chemical restraints can effectively manage these situations without significant safety issues. Moreover, there is little criticism raised against sedating and restraining seriously ill patients in intensive care units. In contrast, criticisms are selectively focused on the use of chemicals on psychiatric patients who appear lucid and have no urgent physical health issues. This becomes particularly problematic when patients are capable of making informed decisions but refuse help. Administering chemical restraints in such cases can be viewed as a violation of basic human rights.

This dilemma is further complicated by the potential long-term psychological impacts of chemical restraints. While they may provide short-term safety, they can erode trust between patients and healthcare providers, potentially harming future therapeutic relationships. Additionally, for patients with histories of trauma or abuse, the experience of chemical restraint may be re-traumatizing, potentially exacerbating their mental health conditions.

Given this context of intense debate, this paper aims to examine the heated debates surrounding chemical restraints and argue for their medical necessity in real clinical settings. We will explore the reasons behind the uniquely harsh criticism against the psychiatric treatment, delving into the historical, philosophical, and social factors that led to this impasse. We'll also briefly examine the burden that ethical principlism has placed

on psychiatrists. Furthermore, this paper will explore current developments in managing patients' agitation and violence, focusing on methods that protect patient safety while minimizing human rights violations. Through this review, we aim to guide healthcare providers toward a more balanced approach.

II. Historical Evolution of Chemical Restraint

According to the U.S. Federal Agency Centers for Medicare and Medicaid Services (CMS), chemical restraint is defined as "any drug that is used for discipline or convenience, and not required to treat medical symptoms." In this sentence, "discipline" refers to any action taken to punish or penalize residents, while "convenience" describes any action taken to manage a patient's behavior with minimal effort by the facility and not in the patient's best interest.[1,2]

From the perspective of mental health workers, this definition carries an excessively negative connotation. Despite the fact that restraint is implemented to ensure the safety of patients and their surroundings—ultimately benefiting patients—the underlying motive is dismissed as mere punishment or a means to reduce staff effort.[3] To understand why such negative prejudice has been solidified against chemical restraint in psychiatry, we must first examine its historical context.

The Legacy of Inhumane Psychiatric Practices in the Past

May 24th was designated as World Schizophrenia Awareness Day to commemorate Philippe Pinel, the director of Salpêtrière Hospital in the 18th century, who ordered the removal of chains that bound patients. As this anecdote illustrates, in the past, physical restraint was the only method to control violent behavior in psychotic patients. Even after the introduction of humanitarian treatment, the methods for handling patients who suddenly exhibited violent behavior were extremely primitive.

Before the introduction of antipsychotic drugs, highly addictive substances like opium and morphine, as well as bromide, hyoscine, and paraldehyde were used. After the development of chloral hydrate in the mid-19th century and barbiturates and phenobarbital in the early 20th century, these were widely used for sedation purposes.[4,5]

The term "chemical restraint" carries historical traces of these cruel and primitive practices.[3] The introduction of antipsychotics in the 1950s dramatically reduced invasive and coercive elements in psychiatric care. However, the 1960s and 1970s witnessed a significant shift, with heightened awareness of patients' human rights. This led to the development of guidelines and policies in the 1980s aimed at limiting the use of restraints,

including chemical ones. While the introduction of atypical antipsychotics seemed to offer a safer alternative for chemical restraint, new side effects and high costs prevented their widespread adoption, leaving human rights issues unresolved. By the 2000s, prompted by the UN Convention on the Rights of Persons with Disabilities, there was a push for more stringent regulations and reporting requirements for the use of restraints in healthcare settings. This shift reflected a growing emphasis on patient rights and safety.[6]

Today, the use of chemical restraint is generally considered a last resort, to be used only when less restrictive interventions have failed and when there is an immediate risk of harm to the patient or others. The focus has shifted towards using the least restrictive interventions possible, with an emphasis on de-escalation techniques and environmental modification.

Rapid Tranquilization and Chemical Restraint

Chemical restraint is often confused with rapid tranquilization. However, there are subtle differences in the underlying concept.[7] The term "rapid tranquilization" began to appear in the early 1970s. A notable reference is a 1971 article by Polak and Laycob[8], which described a case where chlorpromazine was administered to a patient with acute schizophrenia, stabilizing him in 6 hours. As it wasn't uncommon to perform some form of psychotherapy on schizophrenic patients, the goal of rapid tranquilization was to stabilize patients sufficiently for them to engage in psychotherapy.

Some view chemical restraint pejoratively, arguing that it merely exploits the side effects of drugs rather than their therapeutic benefits.[9] In contrast, rapid tranquilization carefully sidestepped the criticism under the pretext of being used for therapeutic effects. The calming effects of antipsychotic drugs were beginning to be attributed to their specific anti-dopaminergic effects rather than non-specific sedative effects. Drug manufacturers began marketing antipsychotics for these targeted effects, a strategy that solidified the role of these drugs in chemical restraint. Yet, ongoing debates persisted about the true nature of these practices: Were they merely "chemical straightjackets," or did they offer specific treatments for underlying conditions?[10]

By the 1980s, haloperidol had become the most frequently used antipsychotic for managing acutely disturbed patients, largely due to its perceived effectiveness and safety. However, concerns about its propensity to induce acute dystonia and akathisia led to the search for alternative medications. This shift reflects a desire to use more safe agents and acknowledges that antipsychotics are not always the best or safest choice.

The early criticism against chemical restraints.

In the 1960s and 1970s, the anti-psychiatry movement, led by figures such as R.D. Laing and Thomas Szasz, began to challenge the fundamental assumptions of psychiatric practice.[10,11] They argued that psychiatric interventions were forms of social control rather than medical treatment. Simultaneously, the civil rights movement in the United States began to influence discussions about patient rights in mental health care.[12] Advocates argued that individuals with mental illness were entitled to the same rights and protections as other citizens, including the right to refuse treatment. This perspective challenged the paternalistic approach that had long dominated psychiatric care and raised questions about the use of chemical restraints without patients' consent.[13]

In the 1970s and 1980s, several landmark legal cases helped to establish and reinforce patients' rights in psychiatric settings. These cases often centered on the right to refuse treatment and the conditions under which involuntary treatment, including chemical restraints, could be administered. For example, the case of Rogers v. Okin in the United States established that involuntarily admitted patients had the right to refuse treatment.[14]

In 1991, the United Nations adopted the Principles for the Protection of Persons with Mental Illness and the Improvement of Mental Health Care. These principles emphasized the importance of patient autonomy, informed consent, and the use of least restrictive treatments. While not specifically addressing chemical restraints, these principles laid the groundwork for more critical examination of coercive practices in psychiatric care. These early social movements set the stage for ongoing debates about the ethics of chemical restraints.[15]

III. Ethical dilemma of Chemical Restraint

Coercive treatment

Coercive treatment refers to any form of intervention that is administered without the patient's explicit consent or against their will. It includes involuntary hospitalization, forced medication, chemical and physical restraints.[16,17] The use of these measures has been justified on the grounds of protecting the patient from harm, or to facilitate treatment when the patient is deemed incapable of making reasonable decisions due to their mental state.

The debate surrounding the chemical restraints is part of a broader debate on the justification of coercive treatment in psychiatric care. Human rights advocates argue that persistent violations of patient rights stem from inadequate legal frameworks governing and limiting coercive treatment.[18,19] They further contend that these issues are exacerbated by psychiatrists' and other healthcare providers' insufficient understanding of bioethical principles and lack of human rights sensitivity.[20,21]

Instead, they advocate for a more patient-centered approach, contending that involving individuals in their treatment decisions, even in emergency situation, can foster better long-term therapeutic outcomes. This strategy, which might include offering choices between observation, isolation, medication, or physical restraint, is thought to build trust between physician and patient while boosting treatment adherence.[22]

UN's Position

In 2017, the United Nations (UN) Special Rapporteur, Dainius Pūras, issued a statement calling for "the abandonment of all practices that fail to respect the rights, will, and preferences of all persons equally, and that lead to power imbalances, stigma, and discrimination in mental health settings." This report, titled "Report of the Special Rapporteur on the right of everyone to the enjoyment of the highest attainable standard of physical and mental health" (A/HRC/35/21) was officially adopted by the CRPD (United Nations Convention on the Rights of Persons with Disabilities) and subsequently became a catalyst for intense debate.[23]

This exemplifies the unwavering position of human rights advocates. The report contends that coercive treatment, frequently rationalized by concepts such as "medical necessity," "dangerousness," and "lack of judgment capacity," is not only subjective but also left entirely to psychiatrists' discretion—thus clearly infringing upon patients' human rights. It further asserted that "the existence of a disability shall in no case justify a deprivation of liberty," which is interpreted as prohibiting the use of mental illness as a justification for restricting liberty.[6] While acknowledging the challenges in eliminating all coercive

treatment, the report emphasized the urgency of eventually ending coercive psychiatric treatment without exception.[24]

However, there was disagreement on how to negotiate between medical necessity with human rights, even within the UN itself.[25] A few years earlier, in 2014, the Human Rights Committee adopted General Comment 35 (GC35), which states:

> The existence of a disability shall not in itself justify a deprivation of liberty but rather any deprivation of liberty must be necessary and proportionate, for the purpose of protecting the individual in question from serious harm or preventing injury to others.

The European Convention on Human Rights (ECHR) also refused to fully accept the CRPD's decision. The ECHR explicitly permits the detention of "persons of unsound mind." This means that coercive measures, including involuntary hospitalization or restraint for psychiatric treatment, do not constitute inhuman treatment if their use is deemed a "therapeutic necessity." The Court thus does not separate the question of detention from treatment without consent, nor does it apply different criteria, such as lack of legal capacity, to the latter. [26–28] However, recent trends indicate a shift towards recognizing patients' rights to refuse forced medication, independent of the legality of involuntary hospitalization. [6]

Both positions share the principle that individuals with mental illnesses or disabilities should not be deprived of their rights and autonomy. However, there is a significant divergence in their interpretations. GC35 and ECHR stipulates that deprivation of liberty can be justified in certain circumstances, whereas A/HRC/35/21 maintains an unwavering stance against accepting such exception. Wayne Martin and Sándor Gurbai from the University of Essex had termed this disagreement as "Geneva Impasse."[28] They argued that this impasse arose from an irreconcilable difference between two distinct approaches to legal interpretation: proportionality and absolute interpretation.

Two approaches for legal interpretation

1. **Proportionality interpretation**

The concept of proportionality is a legal principle that determines an action's legality by assessing the balance between its objective, methods, and consequences.[29] Human rights span a diverse spectrum, including civil, cultural, economic, political, and social rights. Despite their inviolable nature, conflicts may arise between these different categories. This principle suggests that compromising certain rights may be permissible if the action represents a proportionate measure to safeguard one or more fundamental rights.[30] It has

been elevated to to the status of a basic principle of interpretation of ECHR.[31] For instance, ECHR ruled in favor of mandatory COVID-19 vaccination, emphasizing the importance of a balanced approach that protects individual rights while safeguarding public health.

When assessing proportionality in the context of human rights restrictions, several key criteria must be met. First, there must be a legitimate aim, such as safeguarding national security, ensuring public safety, or protecting the rights and freedoms of others. Second, the necessity principle, meaning it should not exceed what is required to achieve the intended goal. Third, the restriction must be commensurate with the threat posed, taking into account both individual impact and broader societal implications. These requirements ensure that any limitation on human rights is justified, appropriate, and minimally invasive.[32]

2. **Absolute interpretation**

Certain actions are considered unethical, regardless of the circumstances. While warfare might justify killing enemies to protect national security, the torture or deliberate mistreatment of prisoners of war is unequivocally condemned. These inviolable human rights must be absolutely protected. The prohibitions of torture and slavery exemplify such rights. In contrast to Europe's emphasis on proportionality, American legalists often rule that basic human rights enshrined in the Constitution cannot be infringed upon on any basis. They insist that without some kind of absolute definition, the entire concept of human rights becomes meaningless.[31]

Returning to the aforementioned examples, it is evident that GC35 and ECHR adopts the proportionality interpretation, while A/HRC/35/21 maintains the absolute interpretation approach. The CRPD considers Article 12 (Equal Recognition before the Law) and Article 14 (Liberty and Security of the Person) as absolute human rights. These provisions are applied to psychiatric patients through the following logic:

Firstly, if a person without mental illness retains the right to refuse coercive treatment, the same right must be conferred on psychatric patients. Such rights should not be denied based on illnesses or disabilities.

Secondly, all forms of coercive treatment inherently restrict the patients' liberty and potentially threaten their physical and mental integrity. Consequently, these practices should be prohibited under Article 14. From a clinician's perspective, this logic doesn't make sense. How can medication, officially approved to be used for specific indications and carefully administered to protect a patient's safety, be construed as a threat to their physical and mental integrity? Yet, this appears to be the prevailing view within the UN

Committee. Such a disconnect between clinical reality and policy interpretation is not just frustrating—it's deeply concerning for those on the front lines of patient care.

Numerous research papers advocating for the abolition of chemical restraints and other forms of coercive treatment consistently report their lack of efficacy in promoting ultimate patient recovery.[33,34] These studies also argue that such interventions increase mental stress among healthcare providers, damage the doctor-patient relationship, and undermine patients' motivation for recovery and long-term outcomes.[35,36] Furthermore, from an absolutist standpoint, those little benefits or drawbacks of chemical restraints are largely irrelevant. Even if healthcare providers' sensitivity for human rights improves and ongoing developments make chemical restraints increasingly non-invasive, effectie and safe, absolutists would still view these practices as illegal and a violation of human rights.

In everyday ER and psychiatric ward, medical staffs strive to resolve emergency situations through non-pharmacological interventions whenever possible. When coercive medication becomes inevitable, it implies that other non-pharmacological means have been deemed ineffective. Concurrently, researchers and pharmaceutical companies constantly try to develop less invasive methods that minimize patient resistance.[37,38] These efforts align with the principle of proportionality, aiming to ensure that therapeutic benefits outweigh any potential infringement on patient autonomy inherent in coercive treatments. However, from the absolute interpretative stance implicit in A/HRC/35/21, these endeavors would be just futile. This interpretation, held by many human rights activists, presents a significant challenge to current clinical approaches.[39]

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IV. Underlying Rationales for the Critiques of Chemical Restraint

(Inviolable) Rights to Refuse Treatment

Human rights can be broadly categorized into two types: the right to demand and the right to refuse. The former raises questions about who is responsible for fulfilling these demands, while the latter concerns whose authority one can resist—highlighting issues of power dynamics.

This debate subtly suggests that certain authorities are exerting undue pressure on vulnerable individuals for the greater social good. While these authorities claim to act in the individual's best interest, they are often accused of protecting the existing social order and maintaining their dominance.[40] Governments, capitalists, and social elites are frequently criticized for such hegemony. Moreover, expert groups—particularly medical doctors who monopolize specialized knowledge—have become vulnerable targets of these critiques.

Human rights advocates argue that these expert groups unjustly decree that people with mental illness are incapable of determining their own good, fueling epistemic injustice that pervades mental health law and policy. According to their logic, this abuse of power silences people with disabilities, rather than listening to their genuine experiences or remedying underlying structural inequality.[41]

Disability and Disease (Disorder or Illness)

As the name CRPD (United Nations Convention on the Rights of Persons with Disabilities) suggests, it is believed that social justice is realized when people with disabilities can fully enjoy their rightful place in society. This stance posits that disability stems from societal structure rather than individual shortcomings. Therefore, it is society that must change, not people with disabilities. This perspective is evident in the distinction between disability and disease (disorder or illness). Disability is not something a person possesses but rather something one encounters when dealing with social environment. Disease, on the other hand, is understood as located in the brain and body and can potentially improve with medical treatment.[42]

While society must make accommodations to ensure the rights of those with disabilities, individuals with diseases have to seek and receive treatment to maintain and improve their own health.[43] Human rights advocates often criticise the concept of mental illness as a narrow, biologically biased view held by myopic experts. Yet, one could also argue that dismissing all scientific truths as mere tools for wielding power is itself an even more biased stance. It's crucial to recognize that not all individuals with mental illnesses are

necessarily disabled.[44] While it may be appropriate to consider chronic, treatment-resistant patients as disabled, those who can recover through optimal treatment might be seen as neglecting their duty if they refuse treatment. For instance, parents of critically ill children who fail to provide prescribed medical care or seek timely treatment are guilty of medical neglect or child abuse.[45] Applying this principle to patients themselves, neglecting one's mental health without sound reason could be regarded as self-directed medical neglect.

The distinction between disease and disability also brings about complicated issue in terms of the duty to protect the health of others. According to the Tuberculosis Prevention Act currently in effect in Korea, when there is a concern that a patient may transmit the bacteria to other people, the head of the local government have to order the patient to be admitted to a medical institution. If the patient refuses, the police must force the patient to receive isolation treatment. Patients, medical staffs, and the policemen who do not comply with this order are subject to fines according to the law.[46] This situation bears a striking resemblance to involuntary admission or coercive treatment for individuals with mental illnesses. However, the Tuberculosis Prevention Act have remained largely unchallenged. This is particularly perplexing, given that tuberculosis patients have also historically endured unfair prejudice and social ostracism.

Perhaps this double standard stems from the fact that tuberculosis is an undeniable disease caused by a clear culprit - the tuberculosis bacterium - whereas mental illness is an invisible and ambiguously defined condition. However, even if there is room for debate, it's hard to deny that major mental illnesses are clearly physical disorders that may improve with treatment.

Humanistic and Social Critiques

1. **Monopolized Knowledge as a Means of Power**

It is undeniable that the history of psychiatry has been marred by ignorance and prejudice, and has been misused as a means of social control. In the 19th century, before the term "psychiatrist" was beginning to be used, mental health professionals were called "alienists." This comes from the idea that patients with mental illness were "alien" and needed to be excluded from society.[47]

As this ignorance began to dissipate through medical advancements, psychiatry hoped to be recognized as a legitimate branch of medicine. However, in the 1960s, when left-wing psychiatrists initiated the anti-psychiatry movement, psychiatry faced particularly fierce criticism. These critics attacked modern science and medicine, especially psychiatry, as a camouflaged version of the oppression once carried out by institutional religions[48–50] They

argued that European church had oppressed the masses to justify the monarchs through the dogma of faith. Now, they believed, the dogma of faith had been replaced by science, and psychiatry in particular was seen as supervising people's thinking, feeling, and behavior, imposing the shackles of what is deemed "normal" and “abnormal”.[51]

These radical idea, grouped under the term "biopolitics," have gained increasing traction in academic and social discourse. The critique of power structures in healthcare and scientific knowledge has become widespread over time.[49] At their core lies a philosophy that views knowledge not as objective truth, but as a tool for wielding power and injustice.[52] Israeli historian Yuval Harari argues that populists rejects the notion of objective truth, asserting instead that everyone possesses "their own truth" to be used as a weapon against rivals. From this perspective, any information or knowledge presented as truth or justice is merely a ploy for power, fabricated by capitalists and colonial elites.[53]

2. **Decolonization and cultural sensitivity**

The original meaning of "decolonization" refers to the process by which formerly colonized countries strive to eliminate the social and cultural remnants of the colonial era, revitalize indigenous cultures, and cultivate national pride.[54] However, the concept has expanded to include a political and academic movements, which challenge the system of male dominance and European white supremacy. Decolonization aims to acknowledge and validate the experiences of marginalized and indigenous people while integrating diverse human perspectives. The neurodiversity movement, for instance, is closely tied to decolonization.[55] It challenges the notion of a single standard of "normality" and opposes "ableism"—the belief that only individuals who can function optimally in society are valuable.[56]

From the decolonizing perspective, the standard of normality they are trying to dismantle has been defined from a European white and male-centric perspective. [57,58] Theorists coined a term “epistemic disobedience” to promote engagement in this violent movement.[59,60] And, in this regard, biological psychiatry, which is faithful to the tradition of Western medicine, is nothing more than an outdated practice that needs to be completely overhauled.[61] Their claim is that if every individual's way of living should be respected, then it should also be accepted as is when people with mental illnesses live peculiar lives, cling to irrational thoughts, and are unable to function in society.

This influential political and academic movement posits that the diverse choices asserted by individuals, regardless of disability or health status, should be accepted without constraint. Consequently, it advocates for the illegalization of coercive treatment and promotes individual autonomy in choosing ways to solve their own problems.[59] This

perspective challenges traditional notions of medical intervention and raises doubts about the legitimacy of modern psychiatric practice.

3. **Against Medicalization**

Medicalization is the process by which non-medical problems become defined and treated as medical problems.[62,63] This term is typically used in a negative context. Sociologists argue that policy makers evade their responsibilities by reducing what are actually societal and political issues to pathological, individual problems. Paula Lantz, a key theorist in this field, defines medicalization as "the expansion of medicine as an institution and the application of a medical perspective to human processes and behaviors".[64]

In reality, it's ambiguous where we should draw the line for what are medical problems or not. For an intoxicated patient brought to the ER for disruptive behavior, it's difficult to decide whether to restraint him/her under the diagnosis of acute alcohol intoxication or send to jail for minor offenses. Patients who have attempted suicide would also pose a dilemma. ER doctor might chemically or even physically restrain them to preserve their life. This decision follows from the reasoning that humans instinctively try to preserve their own lives, and deliberately harming themselves can only be understood as a sign of illness. However, if a patient is truly in a desperate situation and is rationally trying to end his/her life, it's questionable whether the restraint could be justified under the pretext of medical necessity.[65]

Setting aside these ambiguities, sociologists seem to criticize medicalization excessively, often aligning with those who view medicine as a tool for monopolizing knowledge. They argue that public health issues like obesity and diabetes, which disproportionately affect low-income groups, are more reflective of complications of capitalism than purely medical conditions.[66] This perspective extends to mental health, where they contend that rising rates of depression, anxiety, and addiction should be addressed as social issues rather than individual problems treatable through psychiatric intervention.[67,68]

4. **Absolutism in psychiatry ethics**

Moral absolutism holds that most, if not all, actions are inherently right or wrong, regardless of context or consequence. It also asserts that moral principles are universally applicable and inviolable. This concept gained influence through Kant's categorical imperative and has helped uphold universal moral values despite cultural difference. However, principles often clash with the nuances of specific situations, and there's no clear guideline on how flexibly these principles can be applied.[69,70] An absolutist approach, exemplified by the CRPD in A/HRC/35/21, considers any decision violating these principles as unethical, allowing no exceptions. In contrast, a relativist or pragmatist

approach focuses on applying principles to real-world scenarios, rather than insisting on their literal application in every instance.

Healthcare providers working in the field often regard ethical principles as ideal goals to strive for. They acknowledging that, in practice, various aspects must be weighed and balanced to maximize overall benefit. In contrast, moral absolutists argue that principles should be adhered to under all circumstances. Therefore, the rationale behind these firm stances warrants careful examination.

Universal Ideals: Absolutists want to create consistent standards that transcend specific contexts and temporal changes. This approach aims to foster a shared understanding of morality that can be universally applied across diverse situations.[71]

Normative standards: Ethical principles set standards that can serve as benchmarks for evaluating ethical decisions and guide future practices. Abandoning these principles as mere ideal could lead to a "slippery slope", where practical necessities justify morally questionable actions.[72]

Inviolable human rights: Absolutists worry that if these principles are compromised in "exceptional" cases, it could engender a culture where patient rights are disregarded more broadly. Upholding ideal ethical standards can function as a protective measure against inviolable human rights.

Educational and Social Functions: Even though ethical principles may seem detached from real-world complexities, they play an essential role in shaping social attitudes and policies. Adhering to principles and repeatedly educating about them is crucial for instilling values in future professionals and policymakers, who may then be better prepared to grapple with moral dilemmas.

Moral absolutism, which insists on unwavering adherence to ethical principles, often faces criticism for its impracticality. This resistance is particularly evident in Anglo-American traditions, which tend to favor consequentialism and utilitarianism. Contemporary bioethics has largely embraced principlism—a theory that bases moral evaluations on prima facie (non-absolute) principles.[71] This approach represents one step back from absolutism, acknowledging that while these principles are universal, they are not absolute. In real-world scenarios, certain principles may be overridden by more pressing ethical considerations. As a result, even if chemical restraint is administered without a patient's consent, it might not be considered unethical if the principles of beneficence and non-

maleficence take precedence over autonomy and justice.[17] Nonetheless, human rights advocates keep enforcing moral absolutism through legal and institutional mechanisms.

V. Up to Date Development in Pharmacological Management of Acute Agitation and Violence

We have examined the perspectives of human rights advocates who seek to prohibit all forms of coercive treatments based on human rights concerns. While mental health professionals raise valid concerns about this strict approach, the core goal of protecting patients' rights and autonomy remains valid after all. Even though moral absolutists may continue to reject, the development of less invasive and less coercive pharmacological treatments represents significant progress in maintaining ethical psychiatric care.

Alternative Routes of Administration

It is unrealistic to expect an agitated patient to be easily persuaded to take oral medication. On the other hand, coercing parenteral injections raises concerns about human rights violations. Moreover, even if the patient accepted to take oral medication, it would take considerable time for the effects to appear. These circumstances have led to the exploration of alternative routes of administration.

The most practical and well-researched routes are inhalers, nasal delivery, orally disintegrating tablets, syrups, and sublingual formulations. Strictly speaking, orally disintegrating tablets and liquid formulations (syrups) cannot be considered alternative routes. They still have to undergo first-pass effect in the liver after being absorbed through the gastrointestinal tract.[73] Other formulations can be absorbed directly into the bloodstream without undergoing first-pass effect, resulting in faster absorption and higher bioavailability. Additionally, they are less invasive than injections, reducing patient resistance and the risk of accidents. Another advantage is that patients themselves or caregivers can administer these formulations without needing the medical supervision.

Inhalable Loxapine (ADASUEVE™)

Currently, one of the typical antipsychotics, loxapine, are available as inhalable formulation under the name of ADASUVE™. When the patient puts the device in their mouth and inhales, a built-in sensor detects the airflow, rapidly heating a heat source, which vaporizes the coated loxapine. The inhaled loxapine vapor condenses into solid particles about 2μm in diameter and settles in the alveoli. The entire delivery process begins within 1 second after the inhalation.[74]

The efficacy of ADASUVE™ has been demonstrated in three large-scale randomized clinical trials. In a study involving schizophrenia patients,{Allen, 2011 #47} when 5-10mg was administered and evaluated after two hours, there was a significant difference in agitation compared to placebo. In the study by Kwentus et al.[75], comparing ADASUVE™ 5, 10mg, and placebo in patients with bipolar disorder, a significant difference in efficacy began to appear as early as 10 minutes. Based on these results, ADASUVE™ was approved and marketed in the United States and Europe in 2012.

However, it can cause bronchospasms that may lead to severe respiratory distress. In the United States, it is only provided through a special program called REMS (Risk Evaluation and Mitigation Strategies).[74] While one of the advantages of inhalers is that patients can self-administer, ADASUVE™ can only be administered in places with adequate medical facilities. Globally, ADASUVE™ is used as a rapid treatment for acute agitation/excitement in patients with schizophrenia and bipolar disorder, but it has not been widely used due to its side effects and high cost.[38] In Korea, although it acquired approval from the Ministry of Food and Drug Safety in 2019, it has been later withdrawn due to poor sales performance.

Olanzapine intranasal formulation

Although not yet officially approved, two intranasal formulations of olanzapine (INP105 and NRL-4) are under development.[38] As olanzapine injections (not long-acting injectable) have already shown robust efficacy for acute agitation/excitement, manufacturers are working to expand its use through developing special formulations.

INP105 is administered through a technology called POD (precision olfactory delivery). Unlike ADASUVE™ which the patient needs to inhale, this device releases the drug by expelling stored gas, allowing administration even if the patient is uncooperative. Another difference from ADASUVE™ is that while ADASUVE™ is absorbed through the lungs, INP105 is absorbed in the upper nasal epithelium rich in blood vessels. This can lead to considerable variations in bioavailability among individuals.[76]

Another olanzapine formulation is NRL-4. This is a mixture of olanzapine and dodecyl maltoside (DDM), where the latter temporarily loosens the junctions between nasal epithelial cells and stimulates transport mechanisms. DDM has been successfully used in sumatriptan nasal spray (Tosymra™) and diazepam nasal spray (Valtoco™). NRL-4 is administered through a simple disposable device, requiring neither inhalation by the patient nor complex device manipulation.[77]

Sublingual Asenapine

Asenapine is an antipsychotic drug that has been in development since the 1990s, acting on 5-$HT_{2A/D2}$ receptors like other atypical antipsychotics. While its efficacy appeared promising, the issue was that when administered orally, its bioavailability remained below 2%.[78] As a result, the development was focused on creating a sublingual formulation that bypasses the gastrointestinal tract. In 2009, sublingual asenapine received FDA approval for the treatment of acute schizophrenia and bipolar disorder. It's worth noting that while the sublingual tablet is absorbed very quickly, its bioavailability is still relatively low at around 35%.

Initially, it was used for patients who had difficulty swallowing or had gastrointestinal issues. However, it gradually began to be used for acute agitation/excitement, taking advantage of its rapid absorption. Although it hasn't received an indication for acute agitation/excitement, Pratts et al.[79] conducted a placebo-controlled clinical trial in 2014. Significant differences from the placebo group began to appear 15 minutes after administration, and there were almost no adverse reactions reported by patients during the 2-hour observation period.

Dexmedetomidine Sublingual Film

Dexmedetomidine, a highly selective α_2-adrenergic receptor agonist, has recently emerged as a promising treatment for acute agitation and excitement. Initially approved by the FDA in 1999 for sedating patients in intensive care units, it received new FDA approval in 2022 as a sublingual film (IGALMI™). The approved indication specifically targets the treatment of acute agitation and excitement in patients with schizophrenia or bipolar disorder. As a result, clinicians are taking great interest in this formulation.

The efficacy of dexmedetomidine sublingual film has been demonstrated in two controlled clinical trials involving patients with schizophrenia, schizoaffective disorder, and bipolar disorders.[80] When administered as a single dose of 180μg or 120μg, it significantly reduced agitation/excitement compared to placebo, with effects appearing as quickly as 20 minutes after administration.[81] Mild drowsiness was the most common side effect, while other effects such as hypotension, dizziness, and dry mouth were generally mild and infrequent. Since it doesn't act on dopamine receptors, dexmedetomidine doesn't cause extrapyramidal side effects.

Moreover, as a sublingual film, it is expected to increase patient acceptance and significantly reduce the sense of shame patients may experience. With its rapid onset of action, fewer required doses, and superior safety profile, it is considered an ideal treatment option. However, questions remain about the effectiveness of its α_2-adrenergic agonist

action on the underlying symptoms of schizophrenia, and whether its higher price is justified by its improved efficacy and safety.[82]

Using Remote Passive Monitoring to Prevent Violence

No matter how quickly medical staffs respond to emergency situations, intervention typically begins only after an incident has already occurred. Since agitation and violence tend to escalate from minor triggers rather than occurring suddenly, early intervention is of utmost importance.

To predict violent incidents in advance, assessment tools such as the BVC (Brøset Violence Checklist) and DAMA (Dynamic Appraisal of Situational Aggression) have primarily been used. However, there are issues with feasibility and acceptability, as staffs must repeatedly assess multiple patients on a daily basis. Moreover, assessments can vary depending on the evaluator, and it is difficult to ensure reliability in interpretation.[83]

To overcome these limitations, passive remote monitoring technologies have been developed.[84] Using wearable sensors, physiological indicators such as electrodermal activity or heart rate variability can be monitored in real-time without significant burden on the patient. As these indicators change before imminent aggressive behavior, analyzing the collected data in real-time through machine learning or artificial intelligence can help predict impending aggressive behavior.[85,86] While still experimental, these techniques are already being applied in forensic psychiatric settings or clinical practice.[87] In addition to physiological signals, motion sensors and audio recorders can detect patients' aggressive verbal expressions or behaviors in advance.[88]

Passive remote monitoring helps enable early intervention before aggressive behavior worsens, but because these devices collect sensitive physiological data, ensuring data security and privacy is crucial. Patients admitted to psychiatric wards, in particular, are likely to be reluctant to use such devices due to the discomfort of wearing them or paranoid suspicions about monitoring. In the current situation where even CCTV recording requires prior consent, the process of obtaining prior consent for real-time monitoring may not be easy. Additionally, automated data analysis systems are needed for real-time interpretation, but reliable systems are not yet readily available.[89]

VI. Conclusion

Can patients who struggle to control their behavior possess the competency to decide on effective treatment methods? While scholars advocate for prior discussion about all the available options—including physical restraints and involuntary injections—doubts arise as to how many patients would willingly opt for such coercive interventions in advance.

Certain scholars argue that all forms of pharmacological treatment inherently contain elements of coercion and human rights violations, advocating instead for humanistic and psychosocial approaches. Apart from their ineffectiveness in dire conditions, this perspective overlooks the fact that psychological interventions can also contain coercive elements. Offering rewards for behavior modification or persuading patients to take oral medication to avoid injections can be seen as informal coercion.[90] If we were to eliminate any form of coercive treatments, the therapist's role would be reduced to merely providing unconditional validation and emotional support, potentially compromising effective treatment.

It is undeniable that patients with severe mental illness retain their fundamental human rights and autonomy, which must be protected at any cost. Moreover, the importance of patient engagement for successful long-term treatment outcomes cannot be overstated. However, this raises a crucial question: When a patient poses an immediate risk to themselves or others, to what extent should clinical intervention be limited by concerns about human rights concern?

Mental health professionals have embraced human rights principles by eliminating closed wards, minimizing involuntary admissions, and significantly reducing hospitalization periods. They've strived to listen to patients' voices and incorporate them into treatment plans, focusing on long-term compliance and recovery—even at the expense of immediate efficacy. However, being barred from intervening in emergency situations for the sake of protecting patients' human rights is very hard to accept. Moral absolutists fear that allowing exceptions to universal ethical principles will lead down a slippery slope to routine human rights violations. Yet, if they continue to deprecate pharmacological treatments in this way, we might go down a slippery slope where all scientific advancements ends up being dismissed as inhumane and remnants of imperialist colonialism.

The field of psychiatry has witnessed a shift from the traditional paternalistic approach toward a more consumer-oriented model, where patients are encouraged to take greater responsibility for their health decisions. However, this evolution should not be interpreted as a wholesale rejection of paternalistic approaches. Rather than viewing these paradigms as mutually exclusive, it may be more productive to recognize that different situations may

call for different approaches. Likewise, rather than complete abandonment of coercive approaches, a more balanced understanding is required—one that recognizes the complex interplay between patient autonomy, clinical necessity, and the fundamental duty of care. As the field continues to evolve, finding the right balance between these competing priorities remains an essential challenge for mental healthcare practitioners and policymakers alike.[91]

The current effort to eliminate the term "chemical restraints" from medical discourse is merely a form of euphemism. Using a phrase like "pharmacological management of acute agitation and violence" instead of "chemical restraint" may seem to sidestep the core ethical dilemma. Unless human rights advocates' absolutist stance becomes more flexible, no matter how much effort is put into developing safer methods with fewer side effects, they are unlikely to be approved. If this trend continues, clinicians, feeling powerless in the face of criticism from all sides, risk gradually neglecting patients and shifting blame onto others.

This unresolved impasse serves no one—neither the patients whose rights hang in the balance, nor the mental health professionals forced to make split-second decisions in high-pressure environments. Now is the time to ask uncomfortable questions and consider difficult answers. The debate surrounding chemical restraint is not merely an academic exercise; it has real, immediate consequences for everyday psychiatric practice.

Availability of Data and Material

Data sharing not applicable to this article as no datasets were generated or analyzed during the study.

Conflicts of Interest

The authors have no potential conflicts of interest to disclose.

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